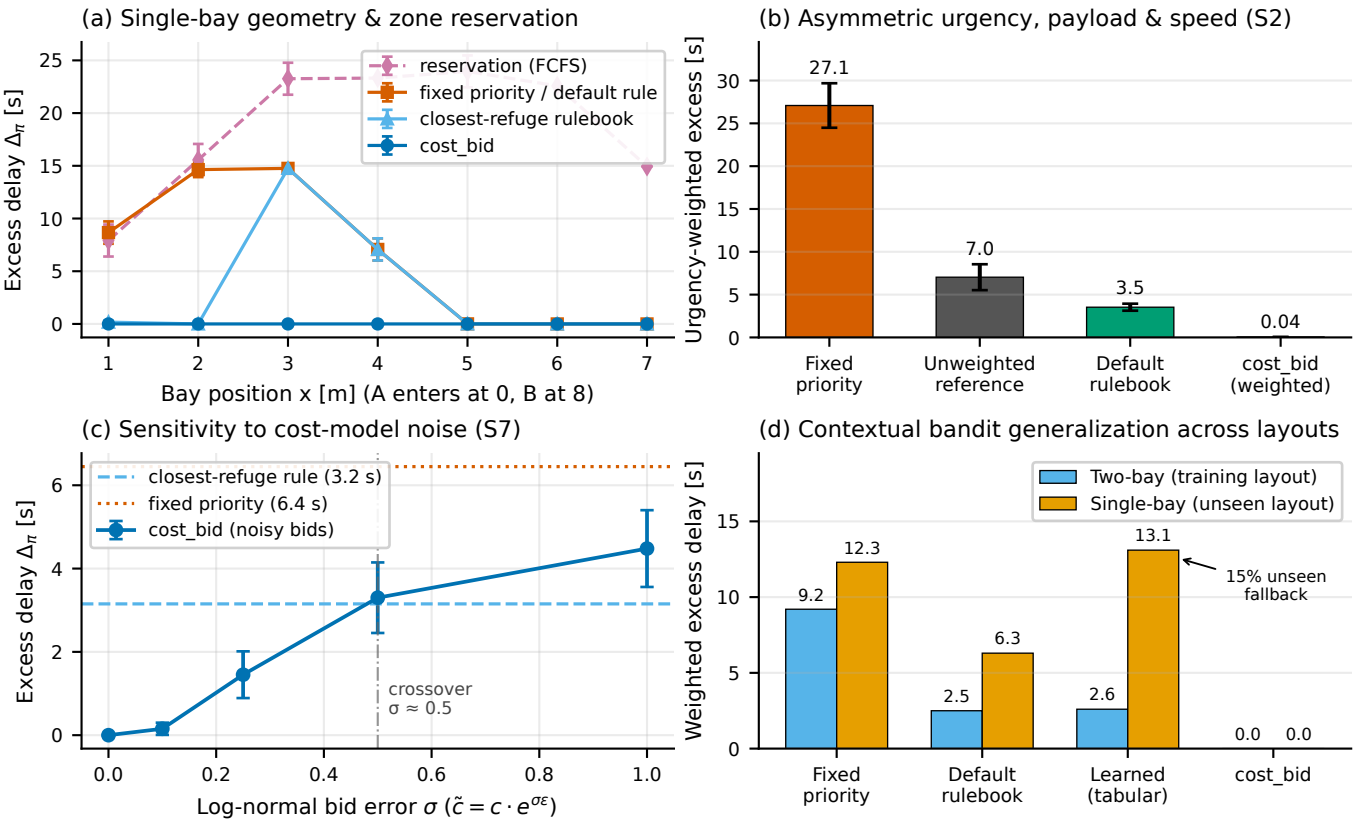
(a) Single-bay geometry & zone reservation
reservation (FCFS)
fixed priority / default rule
closest-refuge rulebook
cost_bid
Excess delay Δ_{π} [s]
Bay position x [m] (A enters at 0, B at 8)
(b) Asymmetric urgency, payload & speed (S2)
Urgency-weighted excess [s]
27.1
7.0
3.5
0.04
Fixed priority
Unweighted reference
Default rulebook
cost_bid (weighted)
(c) Sensitivity to cost-model noise (S7)
closest-refuge rule (3.2 s)
fixed priority (6.4 s)
cost_bid (noisy bids)
crossover
σ ≈ 0.5
Excess delay Δ_{π} [s]
Log-normal bid error σ ($\tilde{c} = c \cdot e^{\sigma\epsilon}$)
(d) Contextual bandit generalization across layouts
Two-bay (training layout)
Single-bay (unseen layout)
Weighted excess delay [s]
9.2
12.3
2.5
6.3
2.6
13.1
0.0
0.0
15% unseen fallback
Fixed priority
Default rulebook
Learned (tabular)
cost_bid